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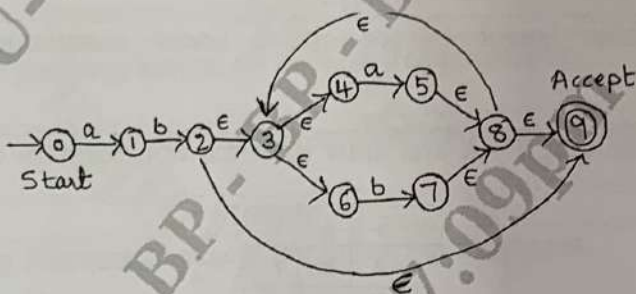
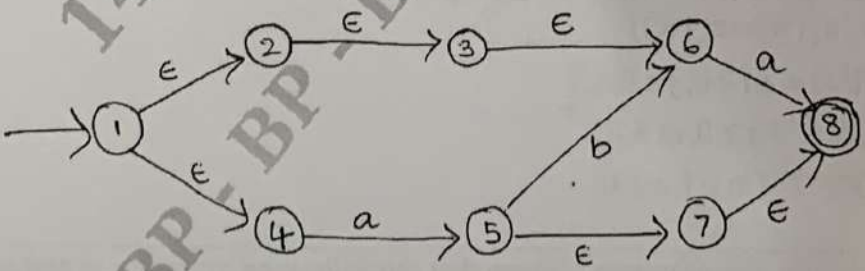
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Fifth Semester B.E/B.Tech. Degree Examination, June/July 2025
Theory of Computation

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
 2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1				M	L	C
1	a.	With a neat diagram, explain a hierarchy of language in automata theory.		5	L2	CO1
	b.	Draw a DFA to accept string of a's and b's ending with ab or ba. Write a transition table for the same.		5	L3	CO1
	c.	Convert the following ϵ -NFA to its equivalent DFA.		10	L3	CO1
 Fig. Q1(c)						
OR						
2	a.	Draw a DFA to accept the languages : i) $L = \{w : n_a(w) \geq 1, n_b(w) = 2\}, \Sigma = \{a, b\}$ ii) $L = \{w : w \bmod 5 \neq 0\}$ on $\Sigma = \{a, b\}$ iii) $L = \{a^p b^q c^r / p \geq 0, q > 0, r \geq 0\}$ on $\Sigma = \{a, b, c\}$.		12	L3	CO1
	b.	Write differences between DFA, NFA and ϵ -NFA.		4	L2	CO1
	c.	Compute ϵ -CLOSURE of each state for the following transition diagram.		4	L1	CO1
 Fig. Q2(c)						

Module - 2

- 3 a. Write regular expression for the following languages :
- $L = \{a^n b^m / n \geq 0, m \geq 0, m + n \text{ is even}\}$
 - $L = \{a^n b^m / n \geq 1, m > 0, nm \geq 3\}$
 - $L = \{a^n b^m / n \geq 4, m \leq 3\}$.

- b. Convert the following automata to a regular expression.

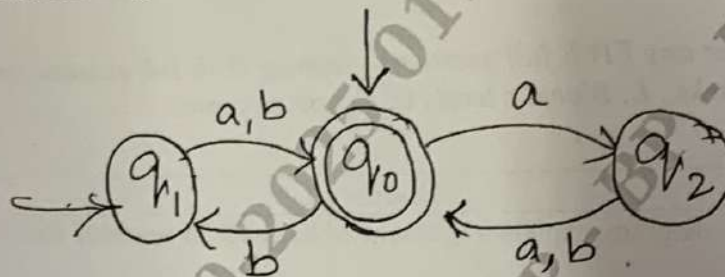


Fig.Q3(b)

- c. Show that $L = \{a^n b^n / n \geq 0\}$ is not regular.

OR

- 4 a. Show that regular languages are closed under complementation and intersection.

- b. Minimize the following DFA and draw the transition diagram of minimized DFA.

δ	a	b
$\rightarrow A$	B	F
B	G	C
* C	A	C
D	C	G
E	H	F
F	C	G
G	G	E
H	G	C

- c. Define Regular Expressions. What are its applications?

Module - 3

- 5 a. Write a grammar for the following languages.

- $L = \{w / n_a(w) = n_b(w)\}$
- $L = \{0^i 1^j / i \neq j, i \geq 0, j \geq 0\}$
- $L = \{a^{n+2} b^m / n \geq 0, m > n\}$
- $L = \{0^m 1^n 2^n / m \geq 1, n \geq 0\}$.

- b. What is ambiguous grammar? Show that the following grammar is ambiguous.
 $E \rightarrow E + E / E * E / id$

OR

6	a.	Explain the following terms : i) Pushdown automata (PDA) ii) Languages of PDA iii) Instantaneous description of a PDA.	6	L1	CO2
	b.	Obtain a PDA to accept $L = \{a^n b^n / n \geq 0\}$, Draw the transition diagram for the constructed PDA.	8	L3	CO3
	c.	What is the difference between deterministic and non-deterministic pushdown automata? Explain with an example.	6	L1	CO3

Module – 4

7	a.	Consider the grammar $S \rightarrow OA / 1B$ $A \rightarrow OAA / 1S / 1$ $B \rightarrow 1BB / OS / O$ Obtain the grammar in CNF.	10	L3	CO3
	b.	Show that $L = \{a^n b^n c^n / n \geq 0\}$ is not context free.	10	L2	CO3

OR

8	a.	State and prove pumping lemma for context free languages.	10	L2	CO3
	b.	Prove that CFL are not closed under complementation and intersection.	10	L2	CO3

Module – 5

9	a.	Design a Turing Machine to accept the following languages. $L = \{0^n 1^n 2^n / n \geq 1\}$.	10	L3	CO3
	b.	Write a short notes on i) Multi – tape Turing Machine ii) Non- deterministic TM.	10	L1	CO4

OR

10		Write a short notes on i) Post correspondence Problem ii) Halting Problem of TM iii) Recursively enumerable language iv) Linear Bounded Automata.	20	L1	CO4
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